

Adi Shankara
INSTITUTE OF ENGINEERING AND TECHNOLOGY
Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
(Owned by Adi Sankara Trust)
DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE)

AIL 411-DEEP LEARNING LAB

PREPARED BY: ASHA ROSE THOMAS

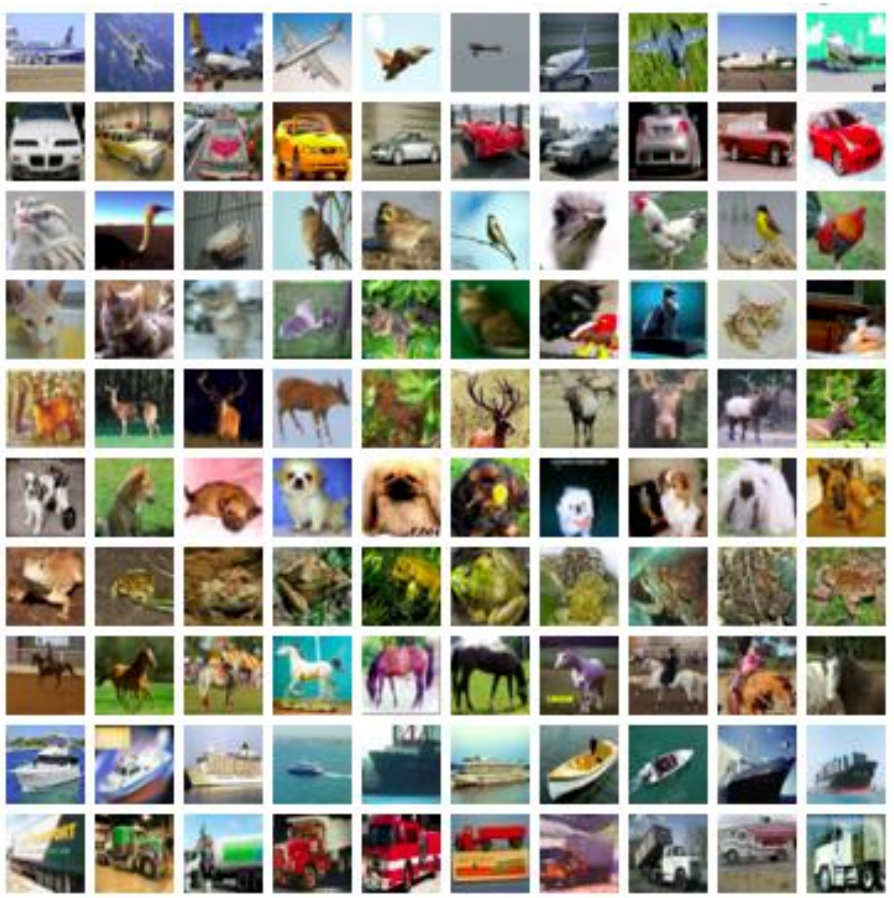
AP,CSE(AI),ASIET,KALADY

EXPERIMENT 3

Implement Feed forward neural network with three hidden layers for classification on CIFAR-10 dataset

CIFAR-10 Dataset

- **CIFAR-10** is a widely used benchmark dataset in computer vision for object recognition tasks.
- Key Characteristics:
  - Image Size: 32x32 pixels
  - Number of Classes: 10 (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)
  - Data Split: 50,000 training images and 10,000 test images
  - Image Format: Color images



The label classes in the dataset

- airplane
- automobile
- bird
- cat
- deer
- dog
- frog
- horse
- ship
- truck

Reference

https://www.cs.toronto.edu/~kriz/cifar.html

https://pyimagesearch.com/2021/05/06/implementing-feedforward-neural-networks-with-keras-and-tensorflow/

https://jovian.com/anurag3301/03-cifar10-feedforward